

The Maximum Number of Intersection Points of Overlapping Regular Polygons

With convexity proofs and a region count

Can Ertürk

An award-winning project at MOSTRATEC (Mostra Internacional de Ciência e Tecnologia)

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Abstract. This note asks how many intersection points can appear at most when several regular polygons are laid over one another in the plane. Because a regular polygon is convex, two regular k -gons can cross in at most $2k$ points, and the proof of this bound is short. For n copies of the same k -gon the count grows to $kn^2 - kn$, and similar closed forms hold for mixtures of polygons of different sizes. The general rule given in the original project is restated here with its index error corrected. The note also settles a question left open earlier: the same arrangement cuts the plane into exactly $kn^2 - kn + 1$ bounded regions. Each formula was checked by a short computer program over many cases.

Keywords: *combinatorial geometry, convex polygons, arrangements, intersection points, Euler's formula.*

1. Introduction

Suppose a few regular polygons are drawn on top of each other in the plane. Their edges cross, and the question of this work is simple to state: at most how many crossing points can there be? Counting problems of this kind sit at the start of combinatorial geometry, and the answers turn out to be useful in tiling, in structural design, and in geometry-based methods for hiding information.

The guiding idea is that the largest count occurs when the polygons are in general position: every pair of them crosses, and no point is shared by three or more polygons at once. When that holds, each pair contributes its own maximum and none of the points coincide, so the totals just add together. The rest of the note makes this precise and proves it.

Compared with the original project, three things are new here. First, the bound for a single pair is proved rather than only observed. Second, the general rule for mixed polygons is corrected. Third, the number of regions the arrangement creates is worked out.

2. How the results were found and checked

The shapes were first drawn in GeoGebra. A second polygon was placed at the same centre and turned by a fraction of the base polygon's symmetry angle, just enough that the edges crossed cleanly without any accidental overlaps. The crossing points

were counted by hand, tabulated, and the pattern across triangle, square and pentagon was generalised by algebra.

For this revision every formula was also tested by a short program that finds all crossings between polygon edges and discards duplicates. It was run on every combination of k in $\{3,4,5,6\}$ and n in $\{2,3,4,5\}$, and the region count was checked the same way with a face count. The program agreed with the formulas in all cases. The code is available with this note.

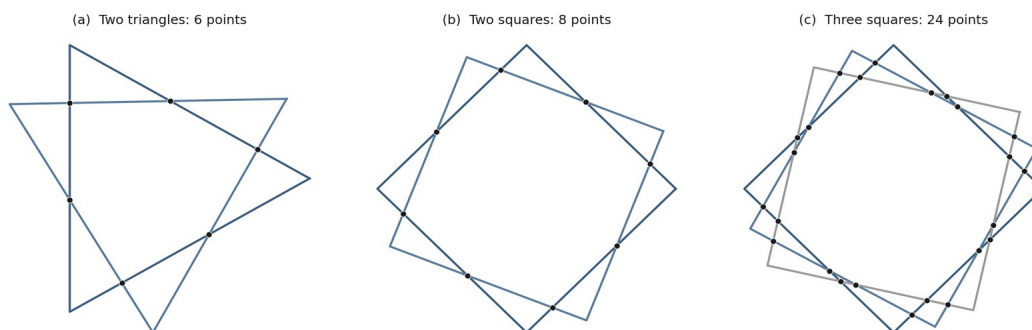


Figure 1. Polygons laid over one another, with the crossing points marked. From left: two triangles give 6 points, two squares give 8, three squares give 24.

3. Two polygons

Theorem 1. *Two regular k -gons cross in at most $2k$ points, and $2k$ can be reached.*

Proof. A regular polygon is convex, so its outline is a convex closed curve. A straight line can enter and leave such a curve only once, so it meets the outline in at most two points. Each of the k edges of the first polygon is a straight segment, so each meets the second polygon's outline at most twice. That gives at most $2k$ crossings in total. Turning one polygon slightly about the shared centre produces a picture in which every edge does cross twice (Figure 2), so the value $2k$ is actually attained. ■

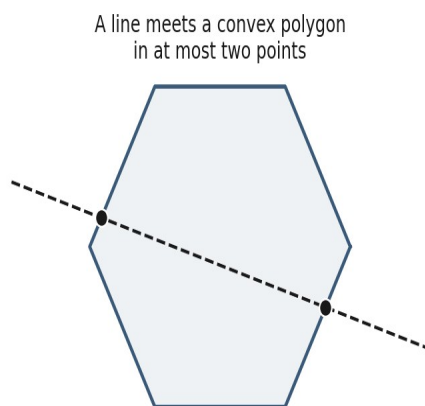


Figure 2. The reason behind the bound: a line meets a convex polygon in two points at most.

A note on the limit. For polygons that are allowed to be non-convex, two simple polygons with m and n sides can cross many more times, up to mn when both side

counts are even. Those configurations need shapes that bend back on themselves. A regular polygon never does, so the convex limit $2k$ is the one that applies here, and Theorem 1 reaches it.

4. Many copies of one polygon

Theorem 2. *For n regular k -gons in general position, the largest number of crossing points is $kn^2 - kn$, equal to $kn(n - 1)$.*

Proof. There are $C(n,2) = n(n-1)/2$ pairs of polygons. By Theorem 1 each pair gives at most $2k$ points, and in general position no two of these points fall on top of each other. Multiplying, the total is $n(n-1)/2$ times $2k$, which is $kn(n-1)$, or $kn^2 - kn$. ■

Putting in the value of k recovers the rules from the original project.

Polygon	Sides k	Most points for n copies
Triangle	3	$3n^2 - 3n$
Square	4	$4n^2 - 4n$
Pentagon	5	$5n^2 - 5n$
Regular k -gon	k	$kn^2 - kn$

Table 1. The rule for n equal polygons and its special cases.

5. Mixing polygons of different sizes

The same pairwise reasoning carries over to polygons of different sizes. Two convex polygons with k_1 and k_2 sides cross in at most $2 \min(k_1, k_2)$ points, since each edge still meets the other outline at most twice and the smaller side count is what limits the total. Two cases come up often, written with the smaller polygon having k sides and k less than n :

$$\text{two } n\text{-gons and one } k\text{-gon: } 2n + 4k \qquad \text{three } n\text{-gons and one } k\text{-gon: } 6(n + k)$$

Take three n -gons and one k -gon. The three n -gon pairs give 3 times $2n$, which is $6n$, and the three n -with- k pairs give 3 times $2k$, which is $6k$. Adding these is $6(n + k)$. The other case follows the same way, and the program confirms both.

5.1 The general rule, corrected

Take a_1 polygons with k_1 sides, a_2 with k_2 sides, and so on, with $k_1 < k_2 < \dots < k_m$. The largest number of crossings is T_1 plus T_2 . Here T_1 counts crossings between polygons of the *same* size, and T_2 counts crossings between polygons of *different* sizes:

$$T_1 = \sum_i k_i (a_i^2 - a_i)$$

$$T_2 = \sum \text{over } i < j \text{ of } 2 \min(k_i, k_j) a_i a_j = \sum \text{over } i < j \text{ of } 2 k_i a_i a_j$$

What was fixed. The original write-up of T_2 mixed up its indices, with terms like $(a_1^2 - a_3)$ that have no clear meaning. The corrected T_2 above simply runs once over each pair of polygon types of sizes $k_i \leq k_j$ and adds $2 \min(k_i, k_j) a_i a_j$, which is $2 k_i a_i a_j$. The full answer is $T_1 + T_2$.

6. Counting the regions

The original project listed, as a question for later, how many *regions* the crossings carve the figure into. That question has a clean answer.

Theorem 3. *For n regular k -gons in general position, the overlaid edges enclose exactly $kn^2 - kn + 1$ bounded regions, which is $P + 1$ where $P = kn^2 - kn$.*

Proof. Read the union of the outlines as a graph drawn in the plane. Its corner points are the kn original polygon vertices together with the P crossing points, so $V = kn + P$. Each crossing splits two edges into two, which adds two segments, so $E = kn + 2P$. For a connected drawing like this, Euler's formula gives the number of enclosed regions as $E - V + 1$, which works out to $(kn + 2P) - (kn + P) + 1$, that is $P + 1$. ■

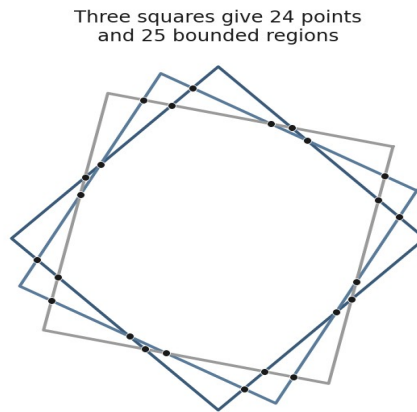


Figure 3. Three squares: 24 crossing points and $24 + 1 = 25$ enclosed regions, as Theorem 3 predicts.

A direct region count by the program matched $P + 1$ in every case tried. This region formula is the main new result of the revision.

7. Summary

- (1) Two regular k -gons cross in at most $2k$ points; the bound follows from convexity.
- (2) n equal k -gons in general position give $kn^2 - kn$ points: triangles $3n^2 - 3n$, squares $4n^2 - 4n$, pentagons $5n^2 - 5n$.
- (3) Two n -gons and one k -gon give $2n + 4k$; three n -gons and one k -gon give $6(n + k)$, for k less than n .
- (4) General mixture: $T_1 + T_2$, with $T_1 = \sum k_i(a_i^2 - a_i)$ and $T_2 = \sum$ over $i < j$ of $2 \min(k_i, k_j) a_i a_j$.

- (5) The enclosed regions number $kn^2 - kn + 1$, one more than the crossing points (Theorem 3).

8. Where this could go next

- Polygons that are not convex, such as stars, where the limit $2k$ no longer holds and the larger mn count becomes possible.
- The same questions in three dimensions, for overlapping regular solids, counting points, edges and regions.
- Colouring the regions or measuring their areas, building on the region count of Section 6.

On the applications side, arrangements like these can serve as keys in geometric transposition ciphers, where a predictable region count gives a predictable block structure. In structural work, knowing where points and regions concentrate can guide where to add material for strength, as in honeycomb layouts, and the closed forms give exact targets for software that optimises a shape automatically.

References

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The figures, table and computer checks in this note were produced by the author using GeoGebra and a Python script.